

1. Limitations of Static Guardrails

Static guardrails don't learn from their own deployment failures



Fixed at Deployment



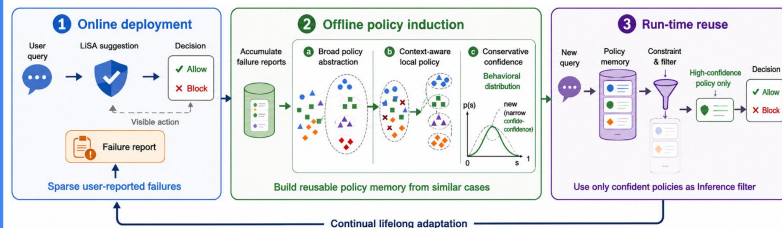
Context-Specific Failures



Expensive to Update

→ **Goal:** adapt guardrails directly from **sparse, noisy** failure reports – without retraining.

2. LiSA Overview



Broad Policies: Generalize individual cases into broad rules.



Local Rules: Prevent overgeneralization in regions with similar semantics but conflicting labels.



Confidence Gating: Only apply high-confidence rules at runtime.

3. Three Mechanisms — each solving one deployment constraint

A

Broad Policy Abstraction

solves SPARSE

Convert sparse failure reports into reusable natural-language policies that generalize beyond individual cases.

$$m = (r, \ell, v)$$

- r : policy statement · ℓ : recommended label · v : support / contradiction counts.
- Broad policy has better safety coverages than individual cases only.

INDUCED BROAD POLICY (example)

“Sharing general or public information is appropriate even for professionals in confidential roles.”

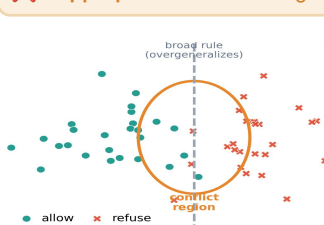
B

Conflict-Aware Local Policy

solves CONSERVATIVE

A single broad policy overgeneralizes when nearby cases split labels. LiSA adds narrow rules only in mixed-label regions.

Example: Sharing Phone Number
 ✓ **Appropriate:** For delivery routing
 ✗ **Inappropriate:** For marketing



C

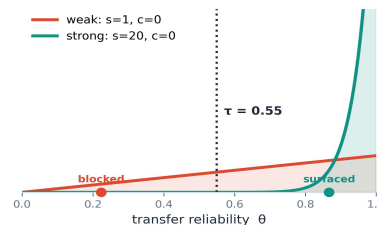
Confidence-Gated Reuse

solves NOISY

Empirical accuracy lies under sparse feedback: 1/1 and 100/100 both look perfect. Gate by the Beta posterior lower bound.

Confidence = Lower Quantile of $\text{Beta}(1+s, 1+c)$ $\geq \tau$

- Reuse scales with evidence volume, not accuracy alone.



4. Results

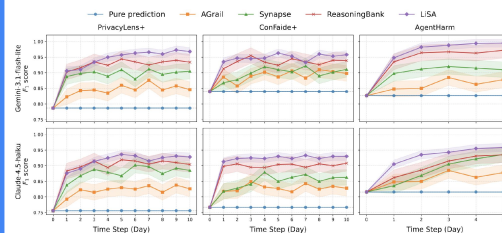


Figure 2 — Evolving F1. Top: Gemini-3.1-flash-lite. Bottom: Claude-Haiku-4.5

Table 2: **Noise robustness.** Final-day F1 score averaged across benchmarks (five seeds) with Gemini-3.1-flash-lite. p : label-flip ratio.

Method	$p=0\%$	$p=10\%$	$p=20\%$
Pure Prediction	0.818	0.818	0.818
AGrail	0.857	0.846	0.832
Synapse	0.891	0.878	0.863
ReasoningBank	0.936	0.882	0.857
LiSA	0.962	0.933	0.917

Table 1: **Ablation study of LiSA.** Final-day macro-F1 and standard deviation across five seeds, averaged across benchmarks with Gemini-3.1-flash-lite.

Ablation	F1
LiSA	0.962 (± 0.013)
w/o Conf gate	0.959 (± 0.025)
w/o Local rules	0.931 (± 0.021)
w/o both	0.925 (± 0.038)

Table 3: **Confidence measurement ablation.** Final-day macro-F1 at $p=20\%$, same setting as left. All variants share LiSA’s two-level memory and differ only in how retrieved broad items are gated.

Confidence	$p=20\%$
No gating	0.854 (± 0.044)
Accuracy $s/(s+c)$	0.883 (± 0.032)
Beta Quantile	0.917 (± 0.025)

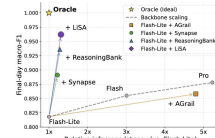


Figure 3: **Latency-F1 trade-off.** Memory-based adaptation pushes the frontier beyond base-model scaling, with LiSA closest to the oracle.

Release

Paper



Contact

